

COCOMO Models

- COCOMO is defined in terms of three different models:
 - the **Basic model**,
 - the **Intermediate model**, and
 - the **Detailed model**.
- The more complex models account for more factors that influence software projects, and make more accurate estimates.

The Development mode

- the most important factors contributing to a project's duration and cost is the Development Mode
 - **Organic Mode:** The project is developed in a familiar, stable environment, and the product is similar to previously developed products. The product is relatively small, and requires little innovation.
 - **Semidetached Mode:** The project's characteristics are intermediate between Organic and Embedded.

The Development mode

- the most important factors contributing to a project's duration and cost is the Development Mode:
 - **Embedded Mode:** The project is characterized by tight, inflexible constraints and interface requirements. An embedded mode project will require a great deal of innovation.

Basic COCOMO model

- Computes software development effort (and cost) as function of program size expressed in estimated lines of code
- Model:

Category	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Basic COCOMO Equations

$$E = a_b k LOC^{b_b}$$

$$D = c_b E^{d_b}$$

where

- E is effort in person-months
- D is development time in months
- kLOC is estimated number of lines of code

$$P=E/D$$

P- Total number of persons
required to accomplish the
project

Merits

- Good for quick, early, rough order of estimates

Limitations:

- Accuracy is limited
- Does not consider certain factors(H/W constraints, personal quality, experience, tools)

Example

- Consider a software project using semi-detached mode with 30000 lines of code. We will obtain estimation for this project as follows:

$$E = a_b k LOC^{bb}$$

- $E = 3.0(30)^{1.12}$
=135 person-month

- Duration estimation

$$D = c_b E^{d_b}$$

$$D = 2.5(135)^{0.35}$$

=14 months

- Person estimation

$$P=E/D$$

$$=135/14$$

$$=10 \text{ persons approximately}$$

Cost Driver Attributes

- Product attributes
 - Required reliability
 - Database size
 - Product complexity
- Computer attributes
 - Execution time constraint
 - Main storage constraint
 - Virtual machine volatility
 - Computer turnaround time

Cost Driver Attributes (Continued)

- Personnel attributes
 - Analyst capability, Programmer capability
 - Applications experience
 - Virtual machine experience
 - Programming language experience
- Project attributes
 - Use of modern programming practices
 - Use of software tools
 - Required development schedule

Intermediate COCOMO Equation

Parameters	a	b	c	d
Organic	3.2	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	2.8	1.20	2.5	0.32

$$E = aikLOC^{bi} \times EAF$$

- where
- E is effort in person-months,
- kLOC is estimated number of lines of code

Merits

- Can be applied to almost entire software for easy and rough cost estimation
- Can be applied at the s/w product component level

Limitations:

Many components difficult to estimate

Software Cost and Effort Estimation

Subtitle: Predicting Project Success

Goal: To accurately predict the **Person-months (Effort)** and **Total Cost** required to develop a software system.

Why Estimate?

- **Budgeting:** Allocate financial resources.
- **Scheduling:** Set realistic deadlines.
- **Bidding:** Determine project price for clients.
- **Risk Management:** Identify potential overruns early.

Software Cost and Effort Estimation

Metric	Abbreviation	Unit	Definition
Size	SLOC	Lines of Code	The total physical or logical size of the final code.
Effort	PM	Person-months	Total labor required for the project (1 PM = 1 person working for 1 month).
Cost	-	Currency (\$)	Total financial expense for development (labor, overhead).
Productivity	-	LOC/PM	Average rate at which a team delivers code.
Quality	D/MLOC	Defects per MLOC	Estimated number of errors remaining in the system.

Software Cost and Effort Estimation

$$\text{Effort (PM)} = \frac{\text{Size (SLOC)}}{\text{Productivity (LOC/PM)}}$$

Estimation Technique: LOC (Line of Code)

Concept: Estimating project size by counting the expected number of **Source Lines of Code (SLOC)**.

The Challenge: How do you estimate the final LOC before writing the code?

Solution: The Three-Point Estimate (PERT-like)

- For each major function, estimate three values:
 1. **Sopt:** Optimistic Size (best case)
 2. **Sm:** Most Likely Size (expected case)
 3. **Spess:** Pessimistic Size (worst case)

Formula for Total Estimated LOC (SLOC_{Total}):

$$\text{SLOC} = \sum \frac{(\text{Sopt} + 4 \times \text{Sm} + \text{Spess})}{6}$$

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Calculating Cost, Effort, & Quality

- 1. Calculate Effort (Person-months):

$$PM = \frac{SLOC_{Total}}{\text{Productivity (LOC/PM)}}$$

- 2. Calculate Total Cost:

$$\text{Total Cost} = PM \times \text{Development Cost per month}$$

- 3. Calculate Cost per LOC (Efficiency):

$$\text{Cost per LOC} = \frac{\text{Total Cost}}{SLOC_{Total}}$$

- 4. Calculate Quality (Estimated Defects):

$$\text{Total Defects} = \frac{SLOC_{Total}}{1,000,000} \times \text{Defect Rate (D/MLOC)}$$

Example Problem: Hostel Management System (H-Mana)

- **Problem Statement:** Create a Student Hostel Management System.
- **Given Metrics:**
 - **Productivity:** 1900 LOC per person-month
 - **Development Cost:** \$6900 per month
 - **Defect Rate:** 1350 D/MLOC
- **LOC Estimates (Weighted Average Method):**

Function	Sopt	Sm	Spess
User interface	1900	2100	4100
Backend development	2600	2700	4600
Testing	2500	1900	3600

Solution: Step 1 - Total LOC (Size)

- Formula: $SLOC = \frac{(S_{opt} + 4 \times S_m + S_{pess})}{6}$

Function	Calculation	Weighted LOC
UI	$(1900 + 4 \times 2100 + 4100)/6$	2700
Backend	$(2600 + 4 \times 2700 + 4600)/6$	3000
Testing	$(2500 + 4 \times 1900 + 3600)/6$	2300
Total SLOC ($SLOC_{Total}$)	$2700 + 3000 + 2300$	8000 LOC

Solution: Step 2 & 3 - Effort and Cost

- 1. Calculate Person-months (Effort):

$$PM = \frac{SLOC_{Total}}{Productivity} = \frac{8000 \text{ LOC}}{1900 \text{ LOC/PM}} \approx \mathbf{4.21} \text{ Person-months}$$

- 2. Calculate Total Cost:

$$\text{Total Cost} = PM \times \text{Monthly Cost} = 4.21 \times \$6900 \approx \mathbf{\$29,049}$$

- 3. Calculate Cost per LOC:

$$\text{Cost per LOC} = \frac{\text{Total Cost}}{SLOC_{Total}} = \frac{\$29,049}{8000 \text{ LOC}} \approx \mathbf{\$3.63} \text{ per LOC}$$

7.2. Solution: Step 4 - Quality

- Calculate Total Defects:

$$\text{Total Defects} = \frac{\text{SLOC}_{\text{Total}}}{1,000,000} \times \text{Defect Rate}$$

$$\text{Total Defects} = \frac{8000}{1,000,000} \times 1350 = 0.008 \times 1350 \approx \mathbf{10.8 \text{ Defects}}$$

- Summary for H-Mana Project:
 - Person-months: 4.21
 - Total Cost: \$29,049
 - Cost per LOC: \$3.63
 - Quality (Defects): ≈ 11

Home Work

Consider a software project for a hospital management system. The software is designed to manage patient records, appointments, and billing. The Lines of Code (LOC) of major software functions identified are:

- Patient registration
- Appointment scheduling
- Billing and invoicing

For patient
registration (LOC):

S_{opt} : 2500
 S_m : 3500
 S_{pess} : 5500

For appointment
scheduling(LOC):

S_{opt} : 4200
 S_m : 6300
 S_{pess} : 8400

For billing and
Invoicing (LOC):

S_{opt} : 1800
 S_m : 2400
 S_{pess} : 4800

In this software project, the productivity rate is estimated at 4000 lines of code (LOC) per person-month, and the development cost is \$6000 per person-month.

Compute the following:

- i. Person required per month.
- ii. Total cost of the software development.
- iii. Cost per LOC.